

Azure HA-DR Terraform Case Study

Architecture Overview

Designed and implemented a production-grade, multi-region High Availability (HA) and Disaster Recovery (DR) architecture on Microsoft Azure using Terraform modules. The solution ensures business continuity, security, scalability, and minimal downtime through automated infrastructure provisioning and geo-replication strategies.

Problem Statement

- Need for highly available and disaster recovery-ready infrastructure.
- Secure and encrypted cloud services using Customer Managed Keys (CMK).
- Scalable Kubernetes-based workloads across multiple regions.
- Enterprise-grade network security and threat protection.

Solution Implemented

- Primary + DR AKS clusters across regions.
- Geo-replicated Azure PostgreSQL for database redundancy.
- Azure Firewall with IDPS and TLS inspection.
- Azure Front Door with WAF for global traffic management.
- VNet peering, NSGs, UDR for secure networking.
- CMK-enabled AKS, ACR, Storage, and Database services.

Terraform Module Architecture

Infrastructure was designed using reusable, modular Terraform architecture to ensure maintainability, scalability, and clean separation of concerns.

Key Terraform Design Features

- Reusable Terraform modules (network, AKS, database, security).
- Environment-based deployment (dev, staging, prod).
- Remote backend configuration for centralized state management.
- Structured variables and outputs for clean module interaction.
- Infrastructure consistency across regions.
- Scalable design for future expansion.

Outcome

Delivered a secure, highly available, and DR-ready Azure infrastructure capable of handling production workloads with automated provisioning, reduced manual errors, and enterprise-grade security posture.